

LEVELLING

6.1 GENERAL

For the execution of engineering projects, such as highways, railways, canals, tunnels, dams, irrigation and drainage works, the elevations (*cf.*, Sec.1.5) of different points along the alignment of the project are required. *Levelling* is the operation in surveying performed to determine and establish elevations of points on the surface of the earth or beneath the surface of the earth, to determine differences in elevation between points, and to control grades in construction surveys. Success of the engineering projects such as above, depends upon accurate determination of elevations. Levelling is of prime importance to the engineers, both for design as well as for execution of a project.

Levelling is a branch of surveying which deals with measurements in vertical planes.

6.2 DEFINITIONS OF SOME BASIC TERMS

Some of the terms, *e.g.*, level surface, horizontal plane, elevation, vertical angle, etc., have already been defined in Sec.1.5. Definitions of some of the terms used in levelling are given below.

Datum or datum surface: It is a level surface with respect to which the levels of points are measured or referred to. It may be a standard surface like mean sea-level or an arbitrary surface.

Bench Mark (B.M.): A bench mark is a permanent or semi-permanent physical mark of known elevation. It is set as point of reference in order to provide a point of beginning for determining elevations of other points. A bench mark is a bronze plate either in the top of a concrete pillar or in the foundation of a structure. Other locations for bench mark may be the top of a culvert headwall, the top of an anchor bolt, or the top of a spike driven into the base of a tree. The bench marks are established by government agencies for the convenience of the surveyors and civil engineers. The surveyors or civil engineers may also establish temporary bench marks for their own use.

6.3 CURVATURE AND REFRACTION

Since levelling is measured in vertical plane, the effects of curvature of the earth and atmospheric refraction which occur in vertical direction, must be taken into account. The required correction to the observations are either determined by the appropriate calculations or else by the measurement techniques designed to eliminate these effects.

It is evident from the definitions that a horizontal line departs from a level surface because of the curvature of the earth as shown in Fig. 6.1.

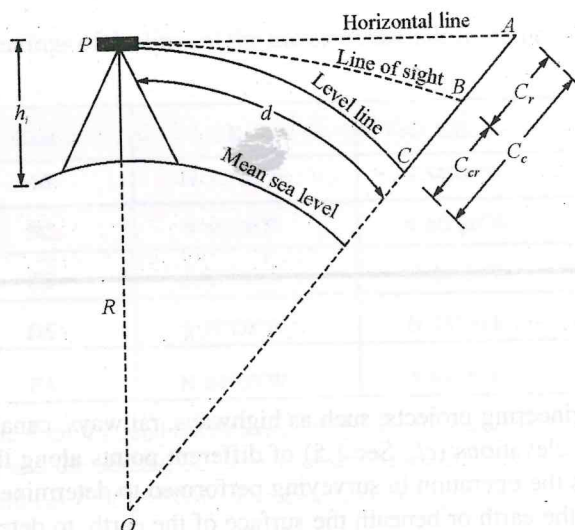


Fig. 6.1 Curvature and refraction

Since the line of sight cannot follow a curved line, an error is introduced and objects appear lower than they really are. This causes increase in reading and, therefore, this correction is negative.

Similarly, the effect of atmospheric refraction is to bend light rays slightly downward as they pass through layers of air of different densities. Light rays reflected from objects and incident on the retina enable us to see them. But the eye perceives them only in the direction tangent to the line of sight at the point of incidence, so the effect of refraction is to make objects appear higher than they really are. Since due to the effect of refraction reading is decreased, the correction for refraction is positive.

In Fig. 6.1, AB is the correction for refraction (C_r) and AC is the correction for earth curvature (C_c). If the distance from the point of tangency P to the point in question in kilometres is d and the mean earth's radius is R then from the right-angled triangle OPA , we have

$$(R + h_i)^2 + d^2 = (R + h_i + C_c)^2$$

where h_i is the height of the instrument axis above the mean sea level. Whilst the value of R is approximately 6370 km, the value of h_i will be of the order of 1.4 m, and therefore, can be neglected compared to the value of R .

Thus

$$(R + h_i)^2 + d^2 = (R + h_i)^2 + C_c^2 + 2(R + h_i)C_c$$

$$d^2 = C_c^2 + 2(RC_c + h_iC_c)$$

Since C_c is very small, the term C_c^2 and h_iC_c will be very small and can be neglected. Therefore,

$$d^2 = 2RC_c$$

$$\text{or } C_c = -\frac{d^2}{2R} \quad (\text{taking sign of the correction})$$

...(6.1)

Under the normal atmospheric conditions, C_r is one seventh of C_c but of opposite sign, i.e.,

$$C_r = +\frac{1}{7}C_c \quad \dots(6.2)$$

Since the effects of curvature and refraction are in opposite directions to each other, the combined correction $BC = AC - AB$ will be

$$\begin{aligned} C_{cr} &= C_c - \frac{1}{7}C_c \\ &= \frac{6}{7}C_c = \frac{6}{7}\left(-\frac{d^2}{2R}\right) = -\frac{3}{7}\frac{d^2}{R} \end{aligned} \quad \dots(6.3)$$

Taking R as 6370 km, the combined correction is

$$C_{cr} = -0.0673d^2 \quad \dots(6.4)$$

In Eq. (6.4), d is expressed in kilometre and C_{cr} in metre.

In ordinary levelling work, the line of sight is seldom more than 100 m long and the corresponding value of C_{cr} is only 0.7 mm. Thus curvature and refraction are rarely significant except in precise levelling.

Distance to the Visible Horizon

In Fig. 6.2, let P be the point of observation at height h and A be the point on the horizon, i.e., a point where the tangent from P meets the level line. Taking value of R as 6370 km and the mean coefficient of refraction as 0.071 for sights over land, the distance to visible horizon is given by

$$d = 0.0483\sqrt{Rh} \text{ kilometres} \quad \dots(6.5)$$

$$\text{or } d = \sqrt{\frac{h}{0.0673}} \text{ kilometres} \quad \dots(6.6)$$

where h is in metres.

In Eqs. (6.5) and (6.6), both curvature and refraction corrections are taken into account.

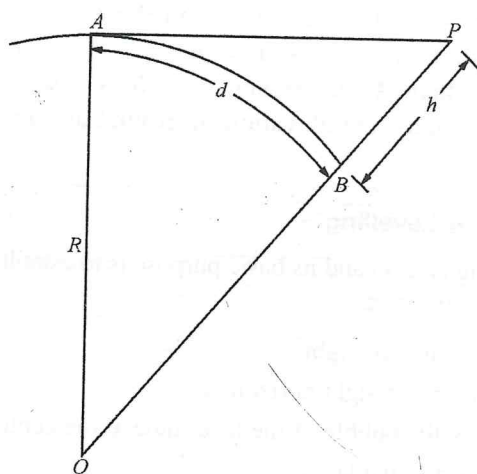


Fig. 6.2 Visible horizon

6.4 METHODS OF LEVELLING

Difference in elevations may be determined by the following methods:

1. Direct differential or spirit levelling
2. Indirect or trigonometric levelling
3. Stadia levelling
4. Barometric levelling.

Direct Differential or Spirit Levelling

Differential levelling or spirit levelling is the most accurate and simple direct method for determining the difference of elevations between two points using an instrument known as *level* with a *levelling staff*. In Fig. 6.3, A is the point of known elevation, say h_A , with respect to which the elevation h_B of the point B is to be determined. The level is set up between the two points and levelled with the help of a level tube attached to the level. When the instrument is levelled, the line of sight would be horizontal. The reading S_A is taken on the levelling staff by first putting it on A and then S_B by putting on B.

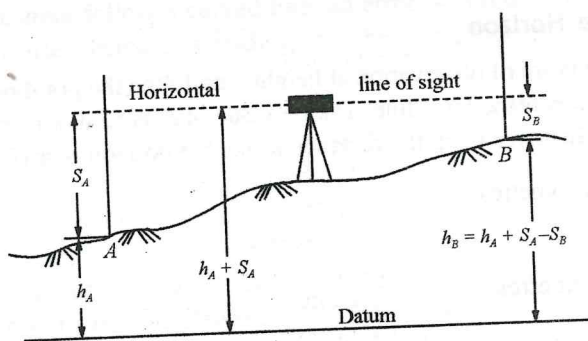


Fig. 6.3 Direct differential levelling

The elevation of the line of sight will be $h_A + S_A$ and the elevation of B will be $h_A + S_A - S_B$. Hence, the difference in elevation $\Delta h_{BA} = h_B - h_A = (h_A + S_A - S_B) - h_A = S_A - S_B$. If $(S_A - S_B)$ is positive, the point B is higher than A and if negative, B is lower than A. It shows that to determine the difference in elevations, the elevation of ground at which the level is set up need not be considered.

Instrument Used in Differential Levelling

The instrument used in levelling is level and its basic purpose is to establish a horizontal line of sight. It consists essentially of the following parts:

1. A telescope which provides a line of sight.
2. A level tube for making the line of sight horizontal.
3. A levelling head for bringing the bubble of the level tube at the centre of its run.
4. A tripod for providing support to the level.

The another instrument which is used with the level to determine amount by which a point is above or below the line of sight, is a levelling staff, straight rectangular rod having graduations and the foot of the staff representing zero reading. The vertical distance between the line of sight and the point over which the staff is held vertically, is the *staff reading*.

Types of levels

Of the various types of levels the following are the most common:

1. Dumpy level
2. Tilting level.

The modern levelling instruments used for precise works or geodetic surveys are:

1. Self-levelling or Automatic level
2. Geodetic level.

The instruments used in reconnaissance surveys to have an idea of change in levels are:

1. Hand level
2. Clinometer.

A brief description of the above levels is given below.

Dumpy level

The dumpy level shown in Fig. 6.4, consists of a telescope, generally the internal focussing type, fixed on a vertical spindle. The telescope tube and the vertical spindle are cast as one piece. The spindle revolves in the socket of the levelling head. The levelling head consists of two parallel plates held apart by three (or four) levelling screws. The upper plate is called tribrach and the lower one is called trivet stage. The telescope can be rotated in the horizontal plane about its vertical axis.

A sensitive level tube is fitted on the top of the telescope or on its side. An inclined mirror (not shown in the figure) is attached to the level tube to enable the observer to view the bubble from the eye end of the telescope without moving round the instrument.

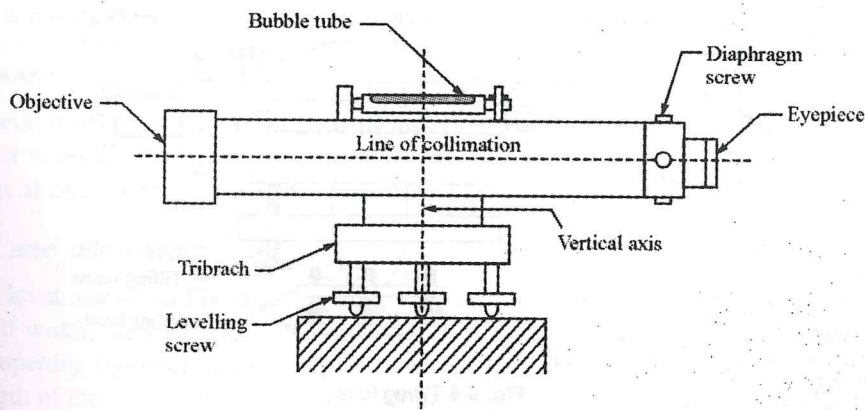


Fig. 6.4 Dumpy level

The cross hairs of the diaphragm normally have a vertical line and a horizontal line. When the instrument is in adjustment, the line of sight of the instrument is perpendicular to the vertical axis and parallel to the bubble tube axis as shown in Fig. 6.5.

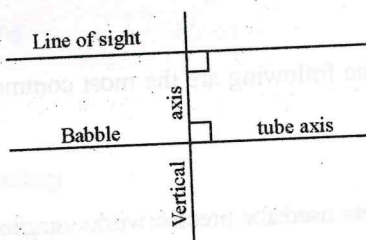


Fig. 6.5 Relationship of the line of sight with the vertical axis and bubble tube axis

The modern dumpy levels are provided with a clamp for clamping the telescope to the tribrach in any desired position. Also a graduated horizontal circle is provided to determine the direction of the telescope.

The dumpy level is simple in construction and requires fewer permanent adjustments. Adjustments once carried out remain for a long period. Before taking reading, the instrument requires temporary adjustments. These are (a) setting up, (b) levelling up, and (c) elimination of parallax, discussed in Sec. 4.4.4. A dumpy level is most useful when from one setting of the instrument, levels of several points are to be determined.

Tilting level

The telescope of a tilting level is not rigidly fixed to the vertical spindle as in the case of dumpy level. The telescope can be tilted on a pivot about a horizontal axis in the vertical plane upwards or downward through a small angle by means of a tilting screw as shown in Fig. 6.6.

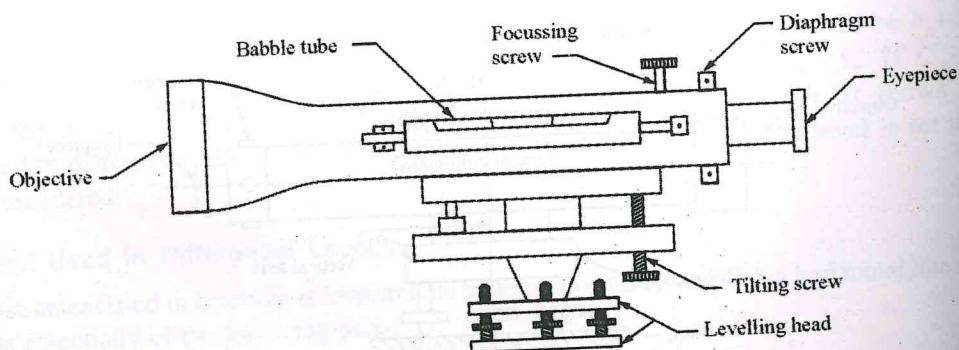


Fig. 6.6 Tilting level

The bull's eye or circular level is fixed to the upper plate of levelling head for approximate levelling by foot screws. The exact levelling of the instrument is done using the tilting screw before taking every reading. The tilting screw is usually graduated to set out gradient lines.

Tilting levels are more robust, compact and accurate than dumpy levels. These have shorter telescope and are lighter. The tilting arrangement saves time required for temporary adjustments. A tilting level is most useful when only a few readings are to be taken from one setting of the instrument.

Automatic level

The automatic level employs a gravity referenced prism or mirror compensator to automatically orient the line of sight (line of collimation). The instrument is quickly levelled when a circular spirit level is used. When the bubble has been centered or nearly so, the compensator takes over and maintains the horizontal line of sight, even if the telescope is slightly tilted. Fig. 6.7 shows the section through an automatic level.

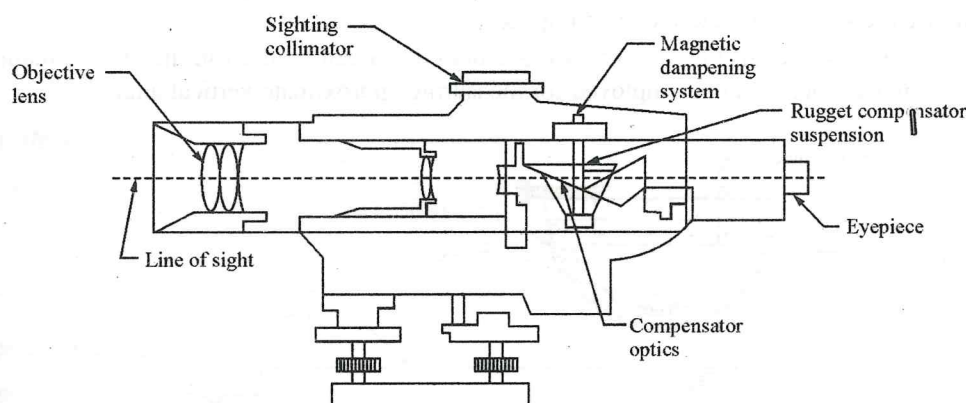


Fig. 6.7 Section through an automatic level

Automatic levels are extremely popular in present-day surveying operations. They are quick to set up, easy to use, and can be obtained for use at almost any required precision.

Geodetic level

The geodetic level is a tilting level used for extreme precision required in first-order levelling. It is made of invar to lessen the effect of temperature and equipped with stadia hairs for determining three staff readings at one observation and for determining lengths of the sights.

Hand level and clinometer

The hand level, shown in Fig. 6.8, is a brass tube with a small level tube mounted on the top. A 45° mirror of half width, on the inside of the main tube, enables the user to view the bubble by reflection through the opening right beneath the bubble tube. As the staff viewed through the level is not magnified, the length of the sight is limited by the visibility of staff readings with the naked eye.

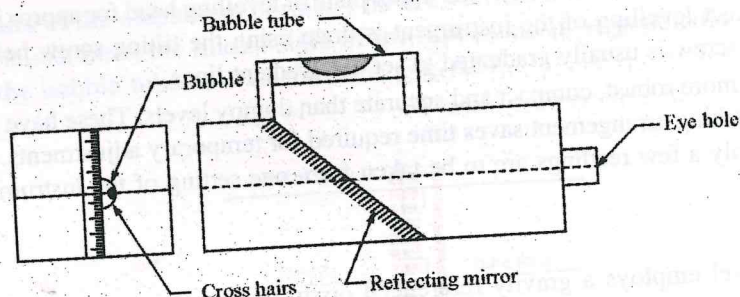


Fig. 6.8 Hand level

The hand level is used in reconnaissance surveys when the extreme accuracy is not required, and in taping to hold the tape horizontally. It is also used for estimating how high or low the dumpy level must be set in order to be able to read the levelling staff.

The clinometer shown in Fig. 6.9, known as Abney level can be used in the similar manner as the hand level. In addition, it can be employed for measuring approximate vertical angles.

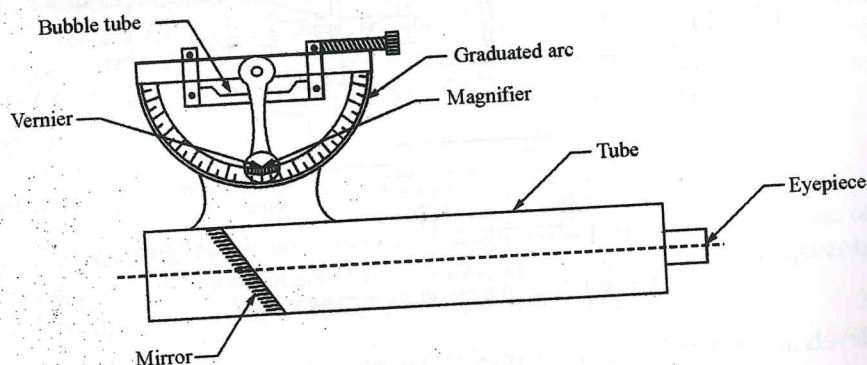


Fig. 6.9 Clinometer

Types of levelling staffs

The levelling staffs are classified as:

1. Self-reading staff, and
2. Target staff.

A self-reading staff is the one which can be read directly by the instrument man through the telescope. On the other hand, a target staff contains a moving target against which the reading is taken by staff man.

Self-reading staff

There are usually three types of self-reading staffs.

Solid staff or single-piece staff

The single-piece or solid staff shown in Fig. 6.10, is a single long flat rod of well-seasoned wood, about 75 mm wide and 25 mm thick, with a metal shoe at the bottom, and graduated on one face. The smallest division is of 5 mm in metric system. The length of the staff is generally 3 m.



Fig. 6.10 Self-reading or solid staff

Folding staff

A folding staff is hinged in the middle so that top half can be folded over to lie against the lower half. When fully extended, the upper part is held firmly in position by a metal pin passing through holes in plates at the side of the hinge. When the staff is not in use it can be folded about the hinge so that it becomes convenient to carry it from one place to the other.

Telescopic staff (sopwith pattern)

A telescopic staff consists of two hollow pieces and one solid top piece, each telescoping into the bottom one. The telescopic staff has advantage over the other types that greater lengths of staff are obtainable up to 4 m to 5 m. When the staff is folded up, it is only a little longer than the lowest piece. When the upper sections are pulled out for use, they are engaged and held by brass spring-catches.

Target staff

A target staff has a sliding target equipped with vernier which can be clamped at any point along the staff. If it is made in more than one piece, separate verniers are provided for each piece.

For taking the reading, the level man directs the staff man to raise or lower the target till it is bisected by the line of sight. The staff man then clamps the target and takes the reading.

Relative merits of self-reading and target staff

- (i) With the self-reading staff, readings can be taken faster than with the target staff.
- (ii) In target staff, adjusting the target is often tedious, and staff man must be a relatively trained person capable of coordinating with the instrument man. In the case of a self-reading type, on the other hand, ordinary man can hold the staff concentrating more on keeping the staff in plumb.
- (iii) The refinement provided by the vernier is often lost when adjusting the target so that it is

bisected by the line of sight. Thus the degree of precision finally attained is not significantly different in the two types. However, a target staff is valuable when very long sights are to be taken.

Classification of Direct Differential Levelling

The direct differential levelling can be further classified into the following methods.

Simple levelling

This is the easiest and straightforward type of direct levelling in which only one setting of the instrument is done to determine the difference in level between two points. It is only suitable when the distance involved is within the sighting range of the level. The instrument need not necessarily lie on the line joining the points.

Differential, compound or continuous levelling

When the two points whose difference in elevation is required are too far apart, or when there are intervening obstacles or the difference in elevation is too large, one setting of the instrument will be inadequate. In such cases, the method of simple levelling is used with as many intermediate level stations as required to reach the forward station maintaining a continuity.

Check levelling

It is employed for the purpose of checking of elevations which have already been determined. Generally check levelling is done at the end of each day's work from last station to the starting station for checking the day's work. Instead of returning to the starting station, sometimes, the day's work is checked by connecting the last station to a point of known elevation or a B.M.

Fly levelling

When a B.M. of known elevation is available, a series of levels may be run along a general direction to reach some other point whose level can be determined and fixed as a temporary B.M., if necessary, the method is called fly levelling. Fly levelling is also employed for the determination of approximate levels of different points. In such cases, the distances as long as possible between the level and staff are measured.

The fly levelling is done where rapidity but low precision is required. It is normally used for reconnaissance surveys or for approximate checking of the levels.

Profile or longitudinal levelling

It is used in route surveys such as a highway, railway, canal, sewer line. The elevations of points along the proposed centre line, at known (usually regular) distance apart, are obtained by differential or fly levelling. The object is to determine the profile of the ground in a vertical longitudinal section (cf., Sec. 6.4).

Cross-section levelling

This type of direct differential levelling is employed to determine the difference of elevations of the ground surface along the lines perpendicular to the centre line of the proposed route. The cross-section levelling is required to determine the configuration across the alignment of the route. Generally, the

profile levelling and cross-section levelling are done simultaneously, to help in estimation of the earthwork involved in a route (cf., Sec. 6.4.10).

Reciprocal levelling

It is the accurate determination of the difference in level between two points by setting up the instrument near each of the points in succession and taking observations of staff readings on both. It is useful when it is not feasible to set up the level midway between the points (cf., Sec. 6.4).

Precise levelling

This is a very accurate method of differential levelling for establishing bench marks or where high precision is required. Refine instruments and procedures enhance the accuracy of the work. It is a specialised form of levelling and is usually conducted by government agencies like Survey of India.

Booking and Reducing Levels

Before discussing the booking and reducing levels, the following terms associated with differential levelling must be understood.

Station

In levelling, a station is that point where the levelling staff is held and not where the level is set up. It is a point whose elevation is to be determined or the point that is to be established at a given elevation.

Height of instrument (H.I.)

For any set up of the level, the elevation of the line of sight is called height of instrument.

Back sight (B.S.)

It is the first reading taken on the staff after setting up the instrument. Initially it is usually made to some form of bench mark or to the point whose elevation has already been determined.

The back sight is usually the first reading taken after setting up the instrument to determine the height of instrument. Thus, the height of instrument is equal to the elevation of the station where staff is held plus the reading on the staff.

Fore sight (F.S.)

It is the reading taken on the staff either held at the last point whose elevation is required or held at the turning point just before shifting the instrument. The fore sight is used to determine the elevation of the staff station. The elevation of the staff station is equal to the height of instrument minus the fore sight.

Change point (C.P.)

A change point or turning point (T.P.) is the point where staff is held and after taking fore sight to determine the elevation of the point where the staff is held, the instrument is shifted and back sight is taken at the same point to determine the new height of instrument. Therefore, at the change point both the back sight and fore sight are taken.

Intermediate station (I.S.)

It is the reading taken on a staff held at a point whose elevation is required, but is not a change point or the last point. Intermediate sites are taken at the intermediate stations.

In Fig. 6.3, S_A is the back sight and S_B is the fore sight. $h_A + S_A$ is the height of instrument. Fig. 6.11 shows the levelling operation between stations A to C. The elevations of the points a , b , c , and C are required to be determined. A is a bench mark whose elevation is known. The staff is held at A and the level is set up at 1. The first reading taken at A is the back sight. Then the staff is held at a , b , and B, the corresponding readings are the intermediate sights on a and b , and the fore sight on B. Then the instrument is shifted to the position 2, and the first reading taken on the staff at B is the back sight. As at B both the fore sight and back sight readings have been taken, it is the change point. After taking back sight at B, the staff is held at c and C, the corresponding readings being the intermediate sight on c and the fore sight on C.

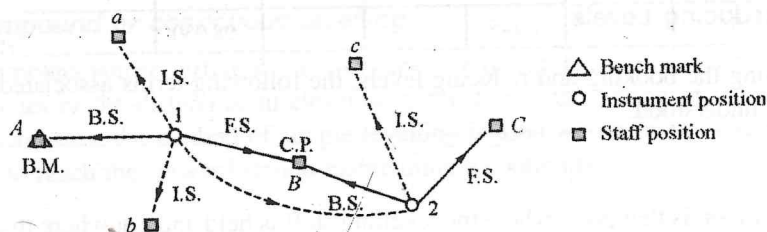


Fig. 6.11 Back sight, fore sight, and intermediate sight

Balancing of sights

When the distances of the points where back sight and fore sight are taken, kept approximately equal, it is known as balancing of sights. This is done to minimise the effect of instrumental and other errors.

Reduced level (R.L.)

Reduced level of a point is the height or depth of the point above or below the assumed datum. It is the elevation of the point. For booking and reducing the levels of points, there are two methods known as (1) Collimation or height of instrument method and (2) Rise and fall method.

Height of instrument method

In this method, height of instrument is calculated for each setting of the instrument by adding the back sight to the known elevation of the point at which back sight has been taken. The reduced level of the points are calculated by subtracting the intermediate sight or fore sight as the case may be, from the height of instrument. The specimen page of a level field book illustrating the method of booking staff readings and reducing the levels of the stations by height of instrument method is shown in Table 6.1.

The station 1 is a bench mark of known elevation of 100.00 m. The staff was held at this station and B.S. 0.685 was taken. Then the staff was held at station 2 and the F.S. 2.905 was taken. Now the instrument was shifted and the B.S. 1.800 was taken at station 2, therefore, this station is a change point. After this two I.S. 1.225 and 2.830 were taken at stations 3 and 4, respectively. The last reading 0.455 for this setting of the instrument is the F.S. at station 5, and then the instrument is shifted to another

position to take the B.S. 2.550 at station 5. Finally the last reading at station 8 is the F.S. 2.625. Between the stations 5 and 8, an I.S. was taken at station 6, and F.S. and B.S. on station 7.

Table 6.1 Height of instrument method of reduction of levels

Station (a)	B.S. (b)	I.S. (c)	F.S. (d)	H.I. (e)	R.L. (f)	Remarks
1	0.685			100.685	100.000	B.M.
2	1.800		2.905	99.580	97.780	C.P.
3		1.225			98.355	
4		2.830			96.750	
5	2.550		0.455	101.675	99.125	C.P.
6		2.745			98.930	
7	2.590		1.555	102.710	100.120	C.P.
8			2.625		100.085	

For the reduction of levels, the steps are given below:

R.L. 100.000 of St.1 + B.S. 0.685 at St. 1	= H.I. 100.685 in Col. (e)
H.I. 100.685 - F.S. 2.905 at St. 2	= R.L. 97.780 of St. 2 in Col. (f)
R.L. 97.780 of St. 2 + B.S. 1.800 at St. 2 (C.P.)	= H.I. 99.580 in Col. (e)
H.I. 99.580 - I.S. 1.225 at St. 3	= R.L. 98.355 of St. 3 in Col. (f)
H.I. 99.580 - I.S. 2.830 at St. 4	= R.L. 96.750 of St. 4 in Col. (f)
H.I. 99.580 - F.S. 0.455 at St. 5	= R.L. 99.125 of St. 5 in Col. (f)
R.L. 99.125 of St. 5 + B.S. 2.550 at St. 5 (C.P.)	= H.I. 101.675 in Col. (e)
H.I. 101.675 - I.S. 2.745 at St. 6	= R.L. 98.930 of St. 6 in Col. (f)
H.I. 101.675 - F.S. 1.555 at St. 7	= R.L. 100.120 of St. 7 in Col. (f)
R.L. 100.120 + B.S. 2.590 at St. 7 (C.P.)	= H.I. 102.710 in Col. (e)
H.I. 102.710 - F.S. 2.625 at St. 8	= R.L. 100.085 of St. 8 in Col. (f)

The height of instrument method provides the following arithmetic check on the reduction of levels:

$$\Sigma B.S. - \Sigma F.S. = \text{Last R.L.} - \text{First R.L.}$$

$$7.625 - 7.540 = 100.085 - 100.000 = 0.085$$

Rise and fall method

In this method, the difference of level between two consecutive points for each setting of the instrument is obtained by comparing their staff readings. The difference indicates a rise if the back sight is more than the fore sight and a fall if it is less than the fore sight. The two consecutive points may have readings as B.S. and F.S., B.S. and I.S., I.S. and I.S., or I.S. and F.S. The rise or fall worked out for all the

points, give the difference in level of each point relative to the preceding one. Therefore, if the R.L. of the preceding point is known then the R.L. of the following point can be obtained by adding the rise or subtracting the fall as the case may be.

Table 6.2 shows a page of the level book illustrating the method of calculating the R.L.'s of stations by the rise and fall method. The booking of readings of B.S., I.S. and F.S. is same as in the height of instrument method.

Table 6.2 Rise and fall method of reduction of levels

Station (a)	B.S. (b)	I.S. (c)	F.S. (d)	Rise (e)	Fall (f)	R.L. (g)	Remarks
1	0.685					100.000	B.M.
2	1.800		2.905		2.220	97.780	C.P.
3		1.225		0.575		98.355	
4		2.830			1.605	96.750	
5	2.550		0.455	2.375		99.125	C.P.
6		2.745			0.195	98.930	
7	2.590		1.555	1.190		100.120	C.P.
8			2.625		0.035	100.085	

For the reduction of levels, let us first determine the rise and fall as below:

B.S. 0.685 at St. 1 – F.S. 2.905 at St. 2	= – 2.220 or fall 2.220 in Col. (f)
B.S. 1.800 at St. 2 – I.S. 1.225 at St. 3	= + 0.575 or rise 0.575 in Col. (e)
I.S. 1.225 at St. 3 – I.S. 2.830 at St. 4	= – 1.605 or fall 1.605 in Col. (f)
I.S. 2.830 at St. 4 – F.S. 0.455 at St. 5	= + 2.375 or rise 2.375 in Col. (e)
B.S. 2.550 at St. 5 – I.S. 2.745 at St. 6	= – 0.195 or fall 0.195 in Col. (f)
I.S. 2.745 at St. 6 – F.S. 1.555 at St. 7	= + 1.190 or rise 1.190 in Col. (e)
B.S. 2.590 at St. 7 – F.S. 2.625 at St. 8	= – 0.035 or fall 0.035 in Col. (f)

Now the R.L.'s of station are calculated as below:

R.L. of St. 1	= 100.000 m (known)	
R.L. of St. 2	= R.L. 100.000 of St. 1 – (fall) 2.220	= 97.780 m
R.L. of St. 3	= R.L. 97.780 of St. 2 + (rise) 0.575	= 98.355 m
R.L. of St. 4	= R.L. 98.355 of St. 3 – (fall) 1.605	= 96.750 m
R.L. of St. 5	= R.L. 96.750 of St. 4 + (rise) 2.375	= 99.125 m
R.L. of St. 6	= R.L. 99.125 of St. 5 – (fall) 0.195	= 98.930 m
R.L. of St. 7	= R.L. 98.930 of St. 6 + (rise) 1.190	= 100.120 m
R.L. of St. 8	= R.L. 100.120 of St. 7 – (fall) 0.035	= 100.085 m.

The rise and fall method of reducing the levels, provides two arithmetic checks as below:

$$\Sigma B.S. - \Sigma F.S. = \Sigma Rise - \Sigma Fall = \text{Last R.L.} - \text{First R.L.}$$

$$\text{i.e., } 7.625 - 7.540 = 4.140 - 4.055 = 100.085 - 100.000 = 0.085$$

Comparison of Height of Instrument Method with Rise and Fall Method

1. The height of instrument method is simple, quicker and less tedious.
2. In the height of instrument method, since the check on the calculations for intermediate sights is not available, the mistakes in their levels pass unnoticed.
3. The rise and fall method though more tedious, provides a full check in the calculations for all sights.
4. The height of instrument method is more suitable in case, where it is required to take a number of readings from the same instrument settings.
5. The rise and fall method should be used for a long run of differential levelling for important works.
6. The rise and fall method should be used when there are a number of important intermediate stations.

ILLUSTRATIVE EXAMPLES

Example 6.1

The following readings were taken with a level and a 4 m staff. Draw up a level book page and reduce the levels by

- (a) the rise and fall method.
- (b) the height of instrument method.

0.683 B.M.(51.362 m), 1.109, 1.838, 3.399, (3.877 and 0.451) C.P., 1.405, 1.896, 2.676, 3.478, (3.999 and 1.834) C.P., 0.649, 1.706.

Solution:

The first reading is on a B.M. of elevation 51.362 m, and, therefore, it is to be entered as back sight. As the fifth station is a change point 3.877 is to be entered as a fore sight and 0.451 as a back sight. The tenth station is also a change point, therefore, 3.999 is a fore sight and 1.834 is a back sight. The reading 1.706 being on the last station, is a fore sight. All remaining readings are the intermediate sights.

The readings and the reduced levels of the stations are tabulated below:

- (a) By rise and fall method

Station	B.S.	I.S.	F.S.	Rise	Fall	R.L.	Remarks
1	0.683					51.362	B.M.
2		1.109			0.426	50.936	
3		1.838			0.729	50.207	
4		3.399			1.561	48.646	
5	0.451		3.877		0.478	48.168	C.P.
6		1.405			0.954	47.214	
7		1.896			0.491	46.723	
8		2.676			0.780	45.943	
9		3.478			0.802	45.141	
10	1.834		3.999		0.521	44.620	C.P.
11		0.649		1.185		45.805	
12			1.706		1.057	44.748	
Σ	2.968		9.582	1.185	7.799		

Arth. Check: $2.968 - 9.582 = 1.185 - 7.799 = 44.748 - 51.362 = -6.614$

(b) By height of instrument method

Station	B.S.	I.S.	F.S.	H.I.	R.L.	Remarks
1	0.683			52.045	51.362	B.M.
2		1.109			50.936	
3		1.838			50.207	
4		3.399			48.646	
5	0.451		3.877	48.168		C.P.
6		1.405			47.214	
7		1.896			46.723	
8		2.676			45.943	
9		3.478			45.141	
10	1.834		3.999	46.454	44.620	C.P.
11		0.649			45.805	
12			1.706		44.748	
Σ	2.968		9.582			

Arth. Check: $2.968 - 9.582 = 44.748 - 51.362 = -6.614$

Example 6.2

It was required to ascertain the elevations of two points *A* and *B* and a line of levels was run from *A* to *B*. The levelling was then continued to bench mark of 77.500 m elevation. The reading obtained being as shown below:

B.S.	I.S.	F.S.	R.L.	Remarks
1.525				<i>A</i>
1.776		0.304		
2.005		1.870		
	2.300			<i>B</i>
0.899		1.011		
1.913		2.784		
		2.113	77.500	B.M.

Solution:

Since in this problem the starting point *A* is not a station of known elevation, to find the R.L.'s of *A* and *B*, the computation are to be done from bottom to top. To find the height of instrument, the fore sight readings will be added to the R.L. of the known point and to determine the R.L. of the previous point, the back sight will be subtracted from the so obtained H.I. as shown in the table below:

Station	B.S.	I.S.	F.S.	H.I.	R.L.	Remarks
<i>A</i>	1.525			78.989	77.464	<i>A</i>
	1.776		0.304	80.461	78.685	
	2.005		1.870		78.591	
<i>B</i>		2.300		80.596	78.296	<i>B</i>
	0.899		1.011	80.484	79.585	
	1.913		2.784	79.613	77.700	
			2.113		77.500	B.M.
Σ	8.118		8.082			

Arth. Check : $8.118 - 8.082 = 77.500 - 77.464 = 0.036$

Example 6.3

The following consecutive readings were taken with a dumpy level and 5 m levelling staff on continuously sloping ground at a common interval of 15 m. The first point is having an elevation of 185.275 m. Rule out a page of level field book and enter the readings. Calculate (a) the reduced levels of the points by rise and fall method, and (b) the gradient of the line joining the first and the last point.

0.415, 1.025, 2.085, 2.925, 3.620, 4.595, 0.715, 2.115, 3.090, 4.405

Solution:

Since the readings were taken on a continuously sloping ground, the readings are continuously increasing up to sixth reading and then suddenly the seventh reading is lesser than the sixth one showing that the sixth point is a change point. Therefore, the seventh reading is back sight. Again from the seventh reading, the readings are continuously increasing. Thus,

the 1st reading is B.S.,

the 2nd, 3rd, 4th and 5th readings are I.S.'s

the 6th reading is F.S.,

the 7th reading is B.S.,

the 8th and 9th readings are I.S.'s, and

the 10th reading is F.S.

The page of the level field book with tabulated readings and calculated R.L.'s is given below:

(a)

Station	B.S.	I.S.	F.S.	Rise	Fall	R.L.	Remarks
1	0.415					185.275	Known elevation
2		1.025			0.610	184.665	
3		2.085			1.060	183.605	
4		2.925			0.840	182.765	
5		3.620			0.695	182.070	
6	0.715		4.595		0.975	181.095	C.P.
7		2.115			1.400	179.695	
8		3.090			0.975	178.720	
9			4.405		1.315	177.405	
Σ	1.130		9.000	0.000	7.870		

Arth. Check : $1.130 - 9.000 = 0.000 - 7.870 = 177.405 - 185.275 = - 7.870$

(b) The gradient of the line

$$= \frac{\text{R.L. of the last point} - \text{R.L. of the first point}}{\text{The length of the line joining the first point to the last point}}$$

$$= \frac{177.405 - 185.275}{15 \times 8}$$

$$= - 0.066$$

$$= 1 \text{ in } 15.25 \text{ (falling).}$$

Example 6.4

A page of level field book is reproduced below in which some readings are missing (marked by *). Complete the page with all arithmetical checks.

Station	B.S.	I.S.	F.S.	Rise	Fall	R.L.	Remarks
1	3.65						
2		*		2.75			
3		2.83					
4		3.64					
5	*		7.42				C.P.
6		12.41			7.32		
7		4.32					
8		3.00					
9		- 6.17					Staff held inverted against ceiling
10	*		*			108.26	C.P.
11			*		1.32		
Σ	17.66				25.93		

Solution:

(i) B.S. (3.65) at St. 1 – I.S.(*) at St. 2 = 2.75 (rise)

$$\begin{aligned} \text{or I.S. at St. 2} &= 3.65 - 2.75 \\ &= 0.90 \end{aligned}$$

(ii) B.S.(*) at St. 5 – I.S. (12.41) of St. 6 = 7.32 (fall)

$$\begin{aligned} \text{or B.S. at St. 5} &= 12.41 + 7.32 \text{ (fall)} \\ &= 12.41 - 7.32 \\ &= 5.09 \end{aligned}$$

(iii) ΣB.S. = B.S. at St. 1 + B.S. at St.5 + B.S. at St. 10

$$\text{or } 17.66 = 3.65 + 5.09 + \text{B.S. (*) at St. 10}$$

$$\begin{aligned} \text{B.S. (*) at St. 10} &= 17.66 - 3.65 - 5.09 \\ &= 8.92 \end{aligned}$$

(iv) Now calculate the rise or fall as the case may be, at the stations from 2 to 9 (cf., the tabulated values)

$$\Sigma \text{Fall} = \text{Fall at St. 10} + \Sigma \text{computed fall at remaining stations.}$$